

Asher Siddiqui

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EDUCATION

FAST-NUCES Bachelor of Science in Software Engineering

Expected Graduation: June 2025

Key Courses: Cloud Security, Applied DevOps, Computer Networks, Database Systems, Design & Analysis of Algorithms, Operating Systems, Data Structures, Object-Oriented Programming

Awards & Honors: Dean's List of Honors (3.5+ SGPA) in Fall 2023, Spring 2024, Fall 2024

SKILLS

Languages: Go, Python, JavaScript, Java, C, HTML/CSS

Tools & Technologies: Git, Docker, Kubernetes, AWS, Linux, Bash, Node.js, Express.js, MongoDB, MySQL, Oracle RDBMS

Familiar With: Spring Boot, Postman, Ansible, Prometheus, Redis, CUDA C/C++

WORK EXPERIENCE

Grayhat Pvt Limited

Software Engineering Intern

Islamabad (Remote)

- Led the development of v0.dev's VS Code extension MVP, managing a 2-person dev team to achieve accelerated project delivery and reduced development time.
- Built an automated developer productivity integration system, synchronizing events across GitHub, Kimai, Activity Watch, and Plane, enabling streamlined workflow tracking and increased developer efficiency.

FEATURED PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | *Kubernetes, Ansible, Prometheus, Falco, Go, Python*

(Ongoing) | [GitHub Link](#)

- Deployed a 3-node botnet infrastructure and a 4-node bare-metal Kubernetes cluster using Ansible to simulate real attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline to log system calls and resource metrics, integrating Falco and Node Exporter, enabling efficient training data generation.
- Developed and deployed simulation scripts using Python on the C&C server and botnet infrastructure, enabling orchestration of controlled DDoS attacks.

Rate Limiter | *Go, Docker, Git* | [GitHub Link](#)

- Engineered a Layer 4 rate limiter supporting Token Bucket and Sliding Window algorithms using Go, preventing API misuse.
- Optimized HTTP responses by adding custom rate-limiting headers, improving client-side request handling and ensuring quota transparency.
- Improved system scalability by leveraging Redis for state management, ensuring low-latency request processing under high traffic loads.

Load Balancer | *Go, Express.js, Docker, Git, HTML* | [GitHub Link](#)

- Designed and implemented a high-throughput HTTP reverse proxy implementing round-robin and least connection algorithms using Go, ensuring traffic delegation to backend servers.
- Increased reliability by integrating health checks, ensuring automatic rerouting of traffic to available servers.
- Containerized the load balancer using Docker, enabling seamless deployment and scalability, and consistent performance across environments.

ADDITIONAL PROJECTS

CUDA-based Rendering of Mandelbrot Set ([GitHub Link](#)): Developed a Mandelbrot Set rendering script using CUDA based GPU processing, achieving high-performance fractal visualization.

Project Management System: Engineered a secure three-tier microservices web application using Node.js, JWT, and a normalized database, enabling efficient project tracking.

Bionic Reader ([GitHub Link](#)): Developed a CLI tool that parses and converts PDFs' text into bionic reading-compatible format, enhancing focus retention.

Assess Buddy ([GitHub Link](#)): Coded a CLI tool to streamline the extraction, compilation, and execution of source code inside compressed folders, using Bash, automating the evaluation of student-submitted assignments.